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1 ONE CHARGER TO CHARGE ALL YOUR DEVICES

Now, there is no need to carry different chargers for different products. This Bhubaneswar startup has launched a universal 65W charger using GaN technology that can charge all your devices—be it your mobile phone or your laptop.



In the tech-centric city of Bengaluru, three forward-thinking professionals, Subhendu S. Satpathy, Syed Athiq, and Sidharth Kumar, united to address the need to carry multiple chargers for various digital devices in their daily lives. They began their journey in 2020 with a clear purpose: to eliminate the clutter and inefficiency of multiple chargers and streamline the way we power our devices.

Driven by this idea, they launched a startup in Bhubaneswar focused on creating a universal charging solution. Their

first product was the brainchild of countless customer interviews and market research. Initially aiming for a 100-watt charger, they realised that a 65-watt charger would best meet consumer needs without sacrificing efficiency or convenience. This insight was timely, coinciding with the European Union's move to standardise chargers—a significant validation of their vision.

The patented intelligent power-sharing (IPS) technology detects which devices are connected and their charge levels. For instance, if a laptop is at 50% and a phone

at 20%, the IPS allocates more power to the phone. This dynamic power allocation ensures efficient charging of all connected devices simultaneously.

They claim that even after two hours of charging, their chargers maintain an average temperature of only 50 degrees Celsius, outperforming competitors. Additionally, they have achieved an average efficiency of 94.8% and a peak efficiency of 96%, ensuring fast and effective charging.

The team utilised gallium nitride (GaN) technology, which is known for its efficiency and ability to handle high power in smaller packages. This choice was instrumental in achieving a power density of 16.6 watts per cubic inch (16.38 cubic cm)—a notable feat in the charger market.

On talking about the design challenges faced, Subhendu says, "Our initial challenge was defining the technical specifications for our product, including efficiency, thermals, and size, aiming for best-in-class design. This process involved starting from scratch with PCB design and manufacturing, leading to collaborations with semiconductor companies like Infineon and Renesas. We faced difficulties in scaling down to small-size PCBs and had to adjust our designs for manufacturability, eventually establishing a resilient supply chain. This allowed us to integrate effectively within the local manufacturing ecosystem and build strong industry partnerships, which were essential for our growth and product development."

The current challenge for the